

Time-of-Flight Laser Distance Sensor (TOF400C/VL53L1X)

Hardware Overview

What is it?

The **TOF400C** is a long-distance ranging module built around the **VL53L1X** chip. It acts as a *LIDAR (Light Detection and Ranging)* system, measuring distances up to 400 cm with mm resolution.

Electrical Explanation

Unlike ultrasonic sensors (e.g. *HC-SR04*) that use sound, this is a *Time-of-Flight (ToF)* optical sensor.

- **The Laser:** It fires fast pulses of invisible, eye-safe 940 nm infrared laser light (Class 1).
- **The Stopwatch:** It uses a highly sensitive *SPAD (Single Photon Avalanche Diode)* array to detect the light bouncing back off an object. By measuring the nanoseconds it took the light to return, it calculates the distance!

Example: If you place your hand 1.5m away from the sensor, the laser pulse travels 3.0 m total. The round time is about 10ns.

Key Hardware Facts:

- **Connection:** I²C Bus (SDA/SCL).
- **Default Address:** 0x29.
- **Voltage:** Operates at 5V.
- **Signal Flow:** Infrared Laser → Target Object → SPAD Receiver → I²C Data → Processor.

Software & Programming

kcomp_TOF400C.h

This library manages an array of tuning parameters and dynamic sensitivity adjustments required by the VL53L1X chip. It handles the laser firing in the background and retrieves the data.

Usage: `#include <kcomp_TOF400C.h>`

The Data Structure

Unlike other sensors that give you a single number (like `int`), the library gives you a `struct` (a package of variables). When you call `readTOF400C()`, you get a package containing all of this:

Field(s)	Type	Description
<code>is_valid</code>	<code>bool</code>	<code>true</code> if a laser pulse has successfully bounced back and the value in <code>distance_mm</code> is valid.
<code>distance_mm</code>	<code>uint16_t</code>	The distance to the target in millimeters (mm).

Function Reference

Function Name	Description & Parameters
<code>bool initTOF400C()</code>	Setup: Starts I ² C, checks the device ID, boots the firmware, and applies the tuning parameters. ▪ Returns <code>true</code> if connected, <code>false</code> if missing.
<code>void setTOF400CMode (TOF400C_Mode_t mode)</code>	Configure: Changes the maximum range of the laser.

	mode: Use TOF400C_MODE_SHORT, TOF400C_MODE_MEDIUM, or TOF400C_MODE_LONG.
<code>tof400c_data_t readTOF400C()</code>	Read Data: Checks if a pulse has returned and provides the <code>tof400c_data_t</code> struct.

⚠ Pitfalls & Troubleshooting

The "20cm Wall" (Optical Crosstalk)

- If your sensor only works from 0–20 cm properly, you are experiencing *Optical Crosstalk*.
The Fix: Remove the black cap over the sensor. The laser is clipping the edge of the plastic.

The "Sunlight Blindness" Problem

- If you take your sensor outside on a sunny day, it might max out at 0 mm or freeze.
The Reason: The sun produces a massive amount of infrared radiation. Outside, the sensor gets overwhelmed by the background infrared light, blinding it.
The Fix: ToF sensors work best indoors. If you must use it outside lower the laser's sensitivity by calling `setTOF400CMode(TOF400C_MODE_SHORT)` in your setup.

The Minimum Distance Blindspot

- The sensor requires a minimum of 40 mm of physical space to work properly.

📄 Sample Program

This program initializes the sensor and continuously prints the distance of the nearest object to the Serial Monitor.

```
#include <kcomp_TOF400C.h>

void setup() {
  Serial.begin(115200);
  delay(1000);

  Serial.println("K-Comp Laser LIDAR System");

  if (!initTOF400C()) { // 1. Initialize the sensor hardware
    Serial.println("Error: TOF400C not found. Check I2C wiring!");
    while(1); // Stop here
  }

  // setTOF400CMode(TOF400C_MODE_SHORT); // Optional: Set to short/medium
}

void loop() {
  // 2. Read the sensor; automatically runs the background state-machine
  tof400c_data_t laser = readTOF400C();

  // 3. Only print if a new pulse has successfully bounced back
  if (laser.is_valid) {
    Serial.print("Distance: "); Serial.print(laser.distance_mm);
    Serial.print(" mm | ");
  }
  // No delay() needed! The is_valid check handles the timing naturally.
}
```

💡 **Try it out:** You can find the full sample program **SimpleLaserDistanceTOF400C** in the K-Comp examples folder!